

Design and Validation of a Thrust Vector Controlled Model Rocket Using PID-Kalman Architecture

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Abstract

This paper presents the design, implementation, and simulation-based validation of a thrust vector controlled model rocket. The system uses a custom two-axis gimbal to deflect motor thrust in real time, replacing passive aerodynamic fins as the sole means of attitude stabilization. A closed-loop PID controller continuously corrects pitch and yaw deviations, while a Kalman filter fuses gyroscope and accelerometer data to produce reliable orientation estimates under vibration and high-acceleration conditions. The flight software is structured as an autonomous state machine, managing all phases of flight from launch detection through apogee identification and recovery. Because powered flight testing was not possible due to airspace restrictions, system performance was evaluated entirely through physics-based simulation, including disturbance rejection analysis and a two-axis Monte Carlo study. These simulations validate the control architecture, achieving a 100% recovery rate across a two-axis Monte Carlo study and defining a maximum recoverable tilt of approximately 12° , the physical boundary of stable flight. The broader aim of this work is to demonstrate that advanced control techniques like thrust vectoring are achievable at the hobby and educational scale, and to provide a foundation that other students can build on.

Keywords: thrust vector control; model rocketry; PID control; Kalman filter; sensor fusion; attitude stabilization; gimbal actuation; flight software; autonomous state machine; Monte Carlo simulation; control authority; inertial measurement unit; finless rocket; aerospace engineering.

append: sentence on
the FHS research project's work
using a
minorly adapted
version of the original
flight code

1 Introduction

Thrust vector control (TVC) is a fundamental technology in modern rocketry. It enables in-flight trajectory correction and attitude stabilization by redirecting the thrust generated by a solid or liquid powered rocket engine on a 2-axis gimbal. Since its development, TVC has played a crucial role in major space launch vehicles, from the Saturn and Space Shuttle programs through to modern vehicles such as SpaceX's Falcon 9 and NASA's Space Launch System (SLS) (1). TVC allows the vectoring and angling of the thrust-producing component in order to counteract aerodynamic instabilities and initial launch trajectory errors.

→ sort of already mentioned here

While TVC is a core part of large-scale rockets, it remains niche and underdeveloped in model rocketry due to its high complexity and the manufacturing precision required, and most model rockets have relied on traditional, passive, aerodynamic stabilization using fins instead. The most notable exception is the hobbyist work of Joe Barnard's BPS.space project, which has demonstrated thrust-vectoring model rockets capable of powered, controlled flight and even propulsive landing (2). This paper builds on that precedent by documenting, in full mathematical and software detail, a TVC system intended specifically as an educational and reproducible reference for other students.

→ This may be a stretch.

The system described here is a compact, low-cost, and fully functional thrust vector control package, integrating a gimbal mount for thrust deflection, a high-speed PID control loop for attitude correction, and a Kalman filter for state estimation during powered flight. The vehicle operates entirely autonomously, from launch until landing, transitioning through predetermined flight states based on real-time sensor data.

The ultimate objective of this research is to demonstrate that TVC can

→ appropriate in this context?

JUST A THOUGHT:

it may be worth adding a short clause about how a lot of figuring this system out was trial & error and that I had to make choices and tradeoffs knowing that I had limited resources/options. Also mention how this project was started out of own initiative? → may not be appropriate for this type of work.

be successfully adapted to small-scale rocketry, and thereby increase the accessibility of controllable, reusable, and autonomously governed model rockets to students and hobbyists. The paper also provides detailed engineering insight into system design, flight software development, and simulation-based validation.

MAYBE also add a piece on how this material should be used to be most effective?

2 System Architecture Overview

This section describes the mechanical, electrical, and computational architecture of the thrust-vector-controlled model rocket. Particular emphasis is placed on component selection, system integration, and the engineering tradeoffs required to achieve reliable closed-loop control within the constraints of a small-scale vehicle.

2.1 Mechanical Airframe & Structural Elements

The rocket airframe was designed around a standard 76 mm (3 in) cardboard body tube manufactured by Estes Rockets. This diameter provided sufficient internal volume for the thrust vectoring gimbal, avionics bay, and recovery system while maintaining a manageable mass and aerodynamic profile. Custom structural components were designed in Fusion 360 and fabricated using fused deposition modeling (FDM) 3D printing with PLA filament.

The nose cone was designed as a hollow, aerodynamically contoured structure to reduce drag during ascent while housing the recovery system. Its geometry was optimized to minimize frontal area and promote stable airflow, while still allowing sufficient internal volume for parachute packing and deployment hardware.

Add detail on how the shape of nosecone was optimized?

The avionics bay serves as the structural and functional core of the vehicle.

It houses the flight computer, inertial measurement unit (IMU), power regulation hardware, and wiring harness. The avionics were enclosed within a protective internal sleeve to shield sensitive electronics from combustion byproducts produced during black powder motor operation, including carbon dioxide, nitrogen, and carbon monoxide. This enclosure also provided structural isolation from exhaust turbulence and vibration.

→ add: inherent to hobby scale, commercially available solid rocket motors like the Estes E12-4.

The thrust assembly consists of the motor mount, retention system, and exhaust shielding designed to interface directly with the TVC gimbal. The motor mount was 3D printed with high infill density to maximize rigidity while minimizing mass and preventing unwanted shifts in the center of mass. An Estes E12-4 solid rocket motor was selected for this prototype due to its predictable thrust curve and compatibility with the airframe scale. The motor mount incorporates mechanical retention screws positioned beneath the solid propellant grain to prevent axial movement during ignition and powered flight. When the gimbal is at its neutral position, the motor centerline is aligned with the rocket's longitudinal axis to avoid introducing unintended moments.

↳ Reiterate this sentence
high infill density
≠
minimized mass.

↳ consistent? reliable?

→ any way to clarify?

Passive aerodynamic fins were intentionally omitted from the design. This decision allowed thrust vector control to serve as the sole stabilization mechanism, more closely modeling real-world TVC applications such as the Falcon 9 first stage. While active or passive fins can provide stabilization at higher airspeeds, their exclusion ensured that all attitude correction authority originated from the gimbal system alone.

2.2 Thrust Vectoring (TVC) System Design

The active stabilization system is centered around a two-axis thrust vector control gimbal that allows the rocket motor to deflect independently in pitch and yaw relative to the vehicle's longitudinal axis. This capability

↳ maybe add a diagram which will allow for clarity for users that are unfamiliar.

enables real-time corrective torque application during powered flight.

The gimbal mechanism consists of a nested, dual-axis structure mounted within the aft section of the airframe. It was designed as a multi-part PLA assembly to maximize available degrees of freedom while maintaining sufficient stiffness under thrust loading. Two MG90S digital micro servos actuate the gimbal, selected for their favorable torque-to-weight ratio, rapid response time, and ease of integration with the flight computer. The manufacturer datasheet rates the MG90S at 0.10 s per 60° under no-load conditions (3); this paper uses a more conservative figure of 0.12 s per 60° throughout, reflecting the additional load imposed by the gimbal mechanism and motor assembly relative to the unloaded datasheet rating.

→ clunky sentence.

The gimbal provides a maximum deflection range of ± 10 degrees in both pitch and yaw. This range was determined through simulation and mechanical testing to provide sufficient corrective authority for a rocket of this size while avoiding excessive coupling effects or mechanical interference. Because corrective torque is directly proportional to the deflection angle, this limit defines the upper bound of achievable control authority during powered flight.

→ max deflection range = 10°
 $\neq$
max recoverable angle = 12°

→ This sentence is complete fluff.

→ expand on this.

2.3 Avionics & Sensor Suite

Vehicle orientation and acceleration are measured using an MPU-6050 inertial measurement unit, which integrates a three-axis gyroscope and three-axis accelerometer into a single module. The gyroscope provides high-rate angular velocity measurements suitable for short-term attitude estimation, while the accelerometer provides absolute orientation reference under low dynamic conditions. Sensor data is transmitted digitally to the flight computer via the I²C protocol. Although the MPU-6050 lacks onboard magnetometer or barometric sensors, its high update rate, affordability, and

→ define.

→ define.

→ explain why that would be useful.

compatibility with embedded development platforms made it suitable for this prototype. Its limitations are addressed through software-based sensor fusion, described in Section 3.2. Flight data is logged using the Teensy 4.1's onboard SD card interface at frequencies between 100 and 200 Hz. Logged parameters include IMU measurements, gimbal deflection angles, system state transitions, and apogee detection events. This high temporal resolution enables detailed post-flight analysis of control system performance and disturbance response. Several onboard LEDs provide real-time visual feedback of system status, including power availability, IMU initialization, and flight mode. These indicators proved useful during ground testing and pre-launch diagnostics. Power is supplied by a two-cell lithium polymer battery regulated to 5 V via a buck converter. The system is designed to handle transient current spikes caused by servo actuation, with additional capacitive smoothing included to mitigate voltage sag. Modular JST and Dupont connectors were used throughout the wiring harness to facilitate rapid component replacement and reduce the risk of permanent damage during testing.

→ Its limitations are described in section 3.2 & are addressed through software-based sensor-fusion.

→ Why is this a range? define.

→ simplify

→ update because not accurate anymore.

→ simplify & define.

2.4 Onboard Flight Computer & Software Stack

The flight computer is built around a Teensy 4.1 microcontroller, selected for its high computational throughput, reliability, and integrated storage capabilities. Its ARM Cortex-M7 processor operates at 600 MHz, providing ample processing headroom for real-time control loops, Kalman filtering, and state machine logic without risk of timing overruns. The board's extensive I/O support allows simultaneous interfacing with the IMU, servos, status indicators, and safety switches.

→ define.

At the highest level, the system transitions between predefined operational modes including pad idle, powered flight, apogee detection, descent, re-

covery, and post-landing (detailed in Section 4 and Appendix A.1).

The flight software is implemented using Arduino-based C++, leveraging standard libraries for I²C communication, SD card access, and servo control, alongside a custom Kalman filter implementation. This development approach combines the accessibility of the Arduino ecosystem with the computational performance required for advanced control and data analysis.

→ define here?

→ maybe not because it is detailed later on.

2.5 Justification for TVC vs. Passive Fins

(such as fins)

Compared to passive aerodynamic stabilization methods, thrust vector control offers significantly greater authority during low-speed and early ascent phases, where aerodynamic forces are minimal. Static fins rely on airflow to generate stabilizing moments and therefore become effective only after sufficient velocity has been achieved. In contrast, thrust vectoring allows corrective forces to be applied immediately after ignition, when disturbances and asymmetries are most likely to grow rapidly.

→ excellent limitations

paragraph but add more on TVC including what role the thrust curve plays in corrective authority.

This capability is particularly important for low-mass rockets, which are highly sensitive to wind, thrust misalignment, and manufacturing tolerances. Small deviations in attitude can quickly compound into large trajectory errors if not corrected early. By actively redirecting thrust, the TVC system provides precise and anticipatory corrections, reducing reliance on passive aerodynamic damping.

→ explain briefly how this is possible through PID controllers.

→ mentioned in section 3.1

Furthermore, the omission of fins results in a vehicle that is inherently aerodynamically unstable, as the center of pressure (CoP) remains forward of the center of mass (CoM) throughout powered flight. This configuration effectively mimics the ‘inverted pendulum’ challenge found in orbital-class boosters, necessitating the active, high-frequency balancing

This paragraph deviates from the tone of the previous two paragraphs (sounds like an add-on).

184 performance that only a TVC system can provide.

185 2.6 System Integration Considerations and Challenges

186 Integrating mechanical, electrical, and computational subsystems into a
187 cohesive thrust-vector-controlled vehicle presents significant engineering
188 challenges. The reliability of the rocket depends not only on the perfor-
189 mance of individual components but also on their interaction under condi-
190 tions of high acceleration, vibration, and thermal stress.

something about each component must work in tandem and cohesively in order for it to work.

191 Thermal protection was a primary concern due to the proximity of avionics
192 to the solid rocket motor. Exhaust gases and particulate matter generated
193 during combustion pose a risk to electronic components. To mitigate this,
194 the motor was positioned within the airframe such that its exhaust plume
195 was directed away from sensitive structures, and a lightweight 3D-printed
196 PLA shield wrapped in aluminum foil was installed to provide thermal
197 isolation.

tone needs correction.

198 Vibration posed another significant challenge, particularly for inertial sen-
199 sors. High-frequency oscillations can introduce noise into IMU measure-
200 ments and degrade attitude estimation. To address this, the IMU was
201 mounted using rubber isolation to attenuate vibration, while the gimbal
202 mount and avionics bay were reinforced with cross-braced supports to re-
203 duce torsional flexing. Wiring harnesses were secured using adhesive and
204 strain relief to prevent movement during launch.

This paragraph reads as very formal which is fine but different than section 2.5 for example.

205 Maintaining alignment between the center of mass (CoM) and center of
206 thrust (CoT) is critical in thrust-vector-controlled systems. Even small
207 offsets can introduce unwanted coupling effects that reduce control ef-
208 fectiveness. The CoM was determined experimentally using a balance
209 method with the fully assembled rocket, including the motor. This mea-

define.

210 surement was supplemented with a method-of-moments calculation based
211 on individual component masses and positions to increase confidence in
212 the result. The equation for calculating the moments was derived from the
213 standard center-of-mass relation:

→ simplify.

$$x_{CoM} = \frac{\sum_i x_i m_i}{\sum_i m_i}$$

214 Because weight is simply mass scaled by the local gravitational accelera-
215 tion g (i.e., $W_i = m_i g$), the same relation can equivalently be expressed in
216 terms of measured component weights, with g cancelling from numerator
217 and denominator:

$$x_{CoM} = \frac{\sum_i x_i W_i}{\sum_i W_i}$$

218 where W_i is the i -th component's weight and x_i is its distance from the
219 fixed reference point. This equivalence allowed the balance-test measure-
220 ment, which directly measures weight, to be cross-checked against the
221 method-of-moments calculation based on component masses, increasing
222 confidence in the result.

expand on this explanation
because things are not
defined well enough.
It does not read well
& the mass to weight
explanation is unnecessary.

223 The center of pressure (CoP) was estimated using OpenRocket (4) sim-
224 ulation software, which applies the Barrowman equations (5) to model
225 aerodynamic forces on slender bodies. Center of pressure is governed by
226 the equation:

$$x_{CoP} = \frac{\sum_i C_{N_i} x_i}{\sum_i C_{N_i}}$$

227 where C_{N_i} is the normal force coefficient for each section, and x_i represents
228 its axial distance from the nose tip. The final configuration showed the CoP
229 positioned approximately 80 mm ahead of the CoM, consistent with the di-
230 vergently unstable, finless configuration described in Section 2.5: with no
231 aerodynamic surfaces to generate a restoring moment, the vehicle's aero-
232 dynamic moment grows with tilt rather than opposing it. OpenRocket's

good content but
wording is too
complex.

output provided both a numerical CoP location and an estimated margin given by:

$$\text{Static Margin} = \frac{x_{CoP} - x_{CoM}}{D}$$

where D is the rocket body diameter (76 mm in this design) and x is measured from the nose tip, increasing toward the tail. With the CoP ahead of (i.e., at a smaller x than) the CoM, this expression yields a negative static margin, the expected and intended signature of an unstable, finless airframe. Propellant burn during powered flight causes the CoM to shift forward as mass is consumed from the aft motor, narrowing the CoP–CoM gap and modestly reducing the magnitude of the instability over the course of the burn, though the configuration remains aerodynamically unstable throughout powered flight. Maintaining adequate thrust-vector control authority relative to this instability, rather than seeking a stable aerodynamic configuration, was therefore essential to flight stability: Section 3.4 quantifies the tilt angle beyond which the TVC system’s corrective torque can no longer overcome this destabilizing aerodynamic moment.

With the structural, aerodynamic, and electronic foundations established, the next phase of the project was centered on developing the closed-loop control system responsible for maintaining rocket stability during powered flight. These physical and simulated findings directly informed the PID tuning process and the Kalman filter parameters, ensuring that the control inputs were responsive and accurate during rapidly changing acceleration and vibrations. The following section details the mathematical framework, algorithmic implementation, and practical tuning procedures that governed the TVC’s behavior in real time.

explain what this means.

maybe talk about the implications of an aerodynamically unstable object.

good conclusion.

3 Control Theory and PID Stabilization

TVC requires real-time orientation correction to counteract disturbances such as wind, manufacturing tolerances, or uneven thrust. This is achieved through a closed-loop control system that combines an IMU, a Kalman filter, and a dual-axis PID controller, continuously adjusting the angle of the motor to correct pitch and yaw deviations.

explain the point of this section more and how to navigate it.

3.1 Orientation Control using PID

At the core of the TVC system is a Proportional–Integral–Derivative (PID) controller, which stabilizes the rocket by adjusting the motor's deflection angle in real time. The controller opposes any external force that pushes the rocket away from a vertical attitude (0° tilt), keeping the thrust aligned with the rocket's center of mass. It does this by minimizing the angular error between the desired vertical orientation and the rocket's measured tilt, using attitude data derived from the IMU. Pitch and yaw are stabilized independently: each axis is treated as a separate control loop, with its own error signal, feedback, and correction command sent to the servo motors.

add how it does that briefly.

very good

The PID control law is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

check this equation should be here?

where:

- $u(t)$: the control output, i.e. the commanded servo deflection angle.
- $e(t)$: the angular error at time t , defined as the difference between the measured attitude and the nominal vertical orientation of 0° .
- K_p , K_i , K_d : the tuned proportional, integral, and derivative gains, respectively.

Each term performs a distinct role. The proportional term (K_p) applies a correction in direct proportion to the current angular error, providing the immediate response to a deviation; if it is set too high, however, it can drive the system into oscillation. The integral term (K_i) accumulates the error over time and removes any steady-state offset, ensuring the rocket returns to exactly vertical rather than settling at a small persistent tilt; an overly large integral gain can cause overshoot or slow oscillatory drift. The derivative term (K_d) responds to the rate at which the error is changing, anticipating where the error is heading and damping the response before it overshoots. This improves stability, but because it acts on the rate of change of a noisy signal, an excessive derivative gain amplifies measurement noise, which is a particular concern given that IMU data is inherently noisy. For the validation simulations presented in Section 6, the integral term was set to zero ($K_i = 0$). Over a burn lasting only a few seconds, the steady-state-error correction the integral term provides is of limited benefit, while the windup risk described above is a genuine concern for a fast, short-duration burn; the simulated controller therefore operates as a pure PD loop. The full PID control law, including integral action and the windup protection described in Appendix A.3, remains implemented in the flight software for use in flight regimes (e.g., longer burns or coast-phase corrections) where steady-state offset correction is more beneficial. Tuning these three gains requires iterative ground and simulated testing to reach stability without excessive oscillation. Initial values can be estimated using the Ziegler–Nichols heuristic method (6). That method is deliberately aggressive and tends to produce high gains, which is not suitable for a vehicle that requires stability and minimal overshoot. These initial values were therefore treated only as a starting point and refined empirically through static-fire testing, confirming that servo commands stayed within their actuation limits and that oscillation was minimized during

→ Can I explore this further? show what it would look like to have oscillating motion. define.

→ mention this is what was said in section 2.5

→ which precludes the need for a Kalman Filter.

→ expanding this would help.

→ Flight regimes refers to actively controlling rocket not in 2 DOF test stand correct this.

→ (k_p, k_i, k_d)

→ explain how this works.

309 rapid attitude correction.

310 The PID model has an important limitation: it assumes a linear relation-
311 ship between the input (servo commands) and the output (angular deflec-
312 tion). During powered flight this relationship is in fact nonlinear, owing
313 to aerodynamic effects, a center of mass that shifts as propellant burns,
314 and particularly the time-varying thrust curve of a solid rocket motor. To
315 keep these nonlinearities from destabilizing the loop, the derivative term
316 can be tuned conservatively, and integral windup protection can be applied
317 so that error does not continue to accumulate once the gimbal has reached
318 its maximum deflection.

→ define.

319 3.2 Sensor Fusion using a Kalman Filter

320 The control loop depends on an accurate, real-time estimate of the rocket's
321 orientation, yet neither sensor on the MPU-6050 can provide one on its
322 own. The gyroscope measures angular velocity, which is accurate over
323 short intervals; orientation, however, is obtained by integrating (contin-
324 uously summing) that velocity over time, so small measurement errors
325 accumulate and the estimate gradually drifts. The accelerometer measures
326 the direction of gravity and therefore provides an absolute reference for
327 the rocket's tilt, but during powered flight its readings are corrupted by the
328 vibration and acceleration produced by the motor. Each sensor is reliable
329 precisely where the other is weak: the gyroscope over short timescales, the
330 accelerometer over long ones. A Kalman filter (7) combines the two into
331 a single orientation estimate that is more reliable than either sensor alone.

→ in order to 'feed' the control algorithm.

→ unnecessary (basic calculus = prerequisite).

→ sentences are crafted almost too well.

332 Before stating the filter equations, it is worth fixing the notation, since the
333 subscripts carry specific meaning. The integer k labels the current control
334 cycle, and $k - 1$ the previous one. A caret denotes an *estimate* rather than
335 a measured or true value, so $\hat{x}$ is the filter's estimate of the tilt angle. A

→ kalman

→ clarifying

→ an estimate of the value at step k only based on $k-1$?

subscript written as $k|k-1$ is read as “the value at step k given only the information available through step $k-1$ ”: it is therefore a prediction, formed before the new measurement at step k is taken into account. Finally, because each axis (pitch and yaw) is estimated independently as a single tilt angle, every quantity below is a single number rather than a matrix, which simplifies the standard Kalman expressions.

clarify this ⇒ I don't think it is worth describing the error covariance matrix but is a question worth asking.

Each cycle proceeds in two steps. In the **prediction** step, the filter advances its previous orientation estimate using the angular velocity reported by the gyroscope:

$$\hat{x}_{k|k-1} = \hat{x}_{k-1} + \omega_k \Delta t$$

where $\hat{x}_{k-1}$ is the orientation estimate carried over from the previous cycle, ω_k is the angular velocity measured by the gyroscope during the current cycle, Δt is the time elapsed since the previous cycle, and $\hat{x}_{k|k-1}$ is the resulting predicted orientation, before any correction. Put simply, the new angle is the previous angle plus the amount the rocket is estimated to have rotated since the last update.

Maybe relate this to some standard kinematic equation?

eg... $v_f = v_i + at$
↳ would this simplify explanation?

In the **update** step, the filter corrects this prediction using the accelerometer, weighting the correction by how much it currently trusts that reading:

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k (z_k - H\hat{x}_{k|k-1})$$

where z_k is the tilt angle derived from the accelerometer during the current cycle, $\hat{x}_k$ is the final, corrected orientation estimate for the cycle, K_k is the Kalman gain (defined below), and H is the observation model, which relates the estimated state to the measured quantity. Because the filter es-

explain these a little better.

357 timates exactly the tilt angle that the accelerometer reports, H equals 1 in
 358 this implementation and can effectively be ignored. The term in parenthe-
 359 ses, $z_k - H\hat{x}_{k|k-1}$, is the difference between what the accelerometer mea-
 360 sures and what the prediction expected, and it serves as the filter's error
 361 signal for the current cycle. $\rightarrow$ error covariance matrix.

362 The Kalman gain K_k determines how strongly that correction is applied: a
 363 large gain pulls the estimate toward the accelerometer, while a small gain
 364 leaves it close to the gyroscope-based prediction. The gain is not fixed but
 365 is recomputed every cycle from the filter's own estimate of its uncertainty:

$$K_k = \frac{P_{k|k-1}H^T}{HP_{k|k-1}H^T + R}$$

366 Here $P_{k|k-1}$ is the *predicted error covariance* — the filter's uncertainty
 367 about its own prediction — and R is the *measurement noise covariance*,
 368 a fixed value representing how noisy the accelerometer is expected to be.
 369 Both are variances: single numbers, in units of angle squared, where a
 370 larger value simply means greater uncertainty. The superscript T denotes
 371 the matrix transpose used in the general formulation; since $H = 1$ here, H
 372 and H^T drop out and the gain reduces to the more transparent form $\rightarrow$ define more clearly.

$$K_k = \frac{P_{k|k-1}}{P_{k|k-1} + R}$$

373 In this form the behavior of the gain is clear. The predicted uncertainty
 374 $P_{k|k-1}$ and the accelerometer noise R compete in the denominator, and
 375 the gain always falls between 0 and 1. When the accelerometer is reli-
 376 able relative to the prediction, R is small, the gain approaches 1, and the
 377 measurement is trusted heavily; when vibration makes the accelerometer
 378 unreliable, R is effectively large, the gain approaches 0, and the filter relies

379 on the gyroscope-based prediction instead.

380 The gain can adapt in this way only because the uncertainty P is itself up-
381 dated every cycle, in two steps that mirror the orientation update. During
382 prediction, integrating the gyroscope adds uncertainty, so P grows by the
383 process noise Q :

This just describes how the certainty in measurements is calculated?

$$P_{k|k-1} = P_{k-1} + Q$$

384 where P_{k-1} is the uncertainty carried over from the previous cycle and Q
385 is a fixed value representing the error the gyroscope introduces each step.
386 After the accelerometer correction is applied, the uncertainty shrinks:

$$P_k = (1 - K_k H) P_{k|k-1}$$

387 where P_k is the updated uncertainty passed forward to the next cycle (again
388 with $H = 1$, so the factor is simply $1 - K_k$). Taken together, the two ori-
389 entation equations and these two covariance equations form a single self-
390 correcting cycle: at every step the filter predicts the new orientation and
391 its uncertainty, measures the disagreement with the accelerometer, and cor-
392 rects both.

393 The two fixed quantities Q and R are what the filter is tuned on, and only
394 their relative size matters. A larger Q tells the filter that its gyroscope-
395 based prediction is less trustworthy, which raises the gain and pulls the
396 estimate toward the accelerometer; a larger R does the opposite, favoring
397 the smoother prediction. For this vehicle, the two were chosen so that the
398 estimate stayed responsive during rapid attitude changes without amplify-
399 ing the accelerometer's vibration noise.

→ If applied correctly,

The result is a smooth orientation estimate that suppresses both the short-term vibration noise of the accelerometer and the long-term drift of the gyroscope, and it is this estimate that the PID controller acts on. One limitation follows directly from the sensor suite: the accelerometer can only sense tilt away from vertical, which fixes the pitch and yaw axes, but it carries no information about rotation about the rocket's longitudinal axis. Because the MPU-6050 has no magnetometer to supply an absolute heading reference (8), that rotation (roll) is tracked by the gyroscope alone and slowly drifts. For the short, vertical, and approximately symmetric flights considered here, uncorrected roll does not affect the rocket's ability to maintain a vertical trajectory, so a magnetometer was not included in the design.

This makes good sense but would be helped if a sentence explaining here.

In simulation, using the MPU-6050's datasheet noise characteristics (Section 6.7), the filter reduced instantaneous tilt noise by roughly 70–80%, which in turn improved the accuracy and responsiveness of the downstream PID loop.

→ due to the integration of small errors in measurement.

3.3 Stability Constraints and PID Gain Tuning Methodology

→ emphasize the significance of this noise reduction, its limitations, and how noise may be further reduced.

Designing a closed-loop control system for a thrust-vector-controlled rocket required careful attention to both control authority and physical response constraints. The controller's ability to stabilize the vehicle depended not only on well-tuned gains but also on the mechanical, aerodynamic, and computational limits of the components selected for this project.

The most significant of these constraints was the servo response speed relative to the rate at which the rocket's instability grows. The MG90S servos used in the gimbal have a rated actuation speed of roughly 0.12 s per 60° of travel (3). A full deflection therefore takes on the order of a tenth

⇒ If the control loop was commanding a rotation which the gimbal could not mechanically reach, there would be a discrepancy between the commanded/desired and actual tilt angles.

of a second, which corresponds to an effective control bandwidth of about 8-10 Hz (since $1/0.12 \text{ s} \approx 8$ actuations per second). The control loop on the Teensy 4.1 microcontroller was run at 150 Hz, far above this actuator bandwidth, so that the controller could detect and begin responding to a deviation well before the servo itself became the limiting factor, preventing small errors from propagating into larger oscillations.

this works in theory because the assumption is that the loop will not command full deflections each time?

The control authority of the system, defined as the total corrective torque available from gimbal deflection, is set by the motor thrust and the moment arm between the thrust line and the center of mass. The corrective moment is:

$$\tau = F_T \cdot r \cdot \sin \delta$$

where F_T is the instantaneous motor thrust, r is the distance from the center of mass to the gimbal pivot, and δ is the gimbal deflection angle. Because torque scales with $\sin \delta$ and the gimbal's range is limited to $\pm 10^\circ$, the maximum corrective torque is bounded by that mechanical range. The PID deflection commands were therefore constrained to the same $\pm 10^\circ$ window, since the controller cannot be allowed to request a correction the gimbal cannot physically produce. Commanding more deflection than the hardware can deliver is known as actuator saturation, and limiting the control output prevents it. At the motor's peak thrust of 33.0 N (Section 6.2), this corresponds to the largest restoring moment the vehicle can generate, a figure quantified across the full burn in Section 6.8.

mechanically

run through the calculation to get the maximum torque?

3.4 Limits of Control Authority and Tilt Recovery

The ability of a thrust-vector-controlled rocket to maintain or recover stable flight depends fundamentally on its control authority: the total correc-

451 tive torque the TVC mechanism can supply relative to the rate at which
452 angular disturbances grow. Once the disturbance exceeds what the control
453 mechanism can counter, the rocket can no longer return to vertical, and
454 the result is a loss of stability and a potential uncontrolled failure. This au-
455 thority is bounded by several factors acting together: servo speed, motor
456 thrust characteristics, aerodynamic forces, structural geometry, and sen-
457 sor latency. The following subsections examine these limits in detail and
458 define the theoretical maximum recoverable tilt angle.

→ confusing sentence.

not written in
my voice.

459 The rotational motion of the rocket about its center of mass is governed by
460 Newton's second law for rotation,

$$I\ddot{\theta} = \tau_{\text{net}} \text{ or more traditionally } \tau_{\text{net}} = I\alpha$$

↳ why is separate notation used?

461 where I is the rocket's moment of inertia about the pitch (or yaw) axis, θ
462 is the tilt angle from vertical, $\ddot{\theta}$ is the resulting angular acceleration, and
463 τ_{net} is the sum of all torques acting on the vehicle. That net torque is the
464 difference between the corrective torque supplied by the gimbal and the
465 disturbance torque that drives the rocket away from vertical:

$$I\ddot{\theta} = \tau_{\text{control}} - \tau_{\text{dist}}(\theta)$$

466 Recovering or holding a stable attitude means keeping $\ddot{\theta}$ directed back
467 toward vertical, which requires the corrective torque to match or exceed
468 the disturbance across the range of tilts the rocket actually reaches. This
469 relationship is the foundation of the analysis that follows: the corrective
470 torque is bounded above by the gimbal's mechanical limit (Section 3.4.1),
471 the disturbance torque grows with tilt (Section 3.4.2), and the angle at
472 which the two become equal marks the edge of recoverable flight.

→ so that the net angular
acceleration is always
guiding the vehicle towards
vertical.

↳ restructure sentence.

3.4.1 Corrective Torque Limitations

The total corrective torque generated by the TVC system is set by the motor thrust, the offset between the gimbal and the center of mass, and the gimbal's maximum deflection angle:

$$\tau_{\max} = F_T \cdot r \cdot \sin(\delta_{\max})$$

where:

- F_T is the motor thrust.
- r is the distance from the center of mass to the gimbal pivot.
- $\delta_{\max}$ is the maximum gimbal deflection angle.

This equation captures the first hard limit of the system. Because $\delta_{\max}$ is fixed by the gimbal's mechanical range, the corrective torque the vehicle can generate at a given thrust is also fixed. As the rocket tilts farther from vertical, it requires progressively more torque to counteract the gravitational and aerodynamic moments acting on it. Once the disturbance torque exceeds $\tau_{\max}$, recovery becomes impossible, no matter how precisely the PID controller is tuned.

Speak about how the thrust curve of a solid rocket motor affects this.

3.4.2 Maximum Recoverable Tilt Angle

The maximum recoverable tilt angle is the largest angular deviation from vertical from which the TVC system can still return the rocket to a stable orientation during powered flight. It is found by comparing the maximum corrective torque available from thrust vectoring against the combined disturbance torques acting on the vehicle. $\Rightarrow \text{clarify} = \tau_{\text{net}}$

The dominant disturbance for this analysis is the destabilizing aerody-

495 namic moment, which grows with angle of attack and airspeed. Two terms
 496 that might appear to belong here do not: gravity acts through the vehicle's
 497 center of mass and therefore produces no net moment about that point, and
 498 the $I\ddot{\theta}$ term in Newton's second law for rotation is the system's own iner-
 499 tial response to net torque, not an independent disturbance. At small tilt
 500 angles, the aerodynamic torque scales approximately linearly with the tilt.

not my tone of voice.

501 At small tilt angles, the aerodynamic and gravitational torques scale ap-
 502 proximately linearly with the tilt. As the tilt increases, however, these mo-
 503 ments grow faster than the available corrective torque, which is capped by
 504 the fixed gimbal limit. Once the disturbance torque exceeds the maximum
 505 corrective torque defined in Section 3.4.1, recovery is no longer possible
 506 even with thrust still available. This defines a recovery boundary:

→ citation or explanation?

$$\tau_{\text{dist}}(\theta) \leq \tau_{\text{max}}$$

→ θ = tilt of rocket & δ = tilt of gimbal?

507 where θ is the rocket's tilt angle from vertical. Beyond this boundary
 508 the system diverges: further corrective commands can no longer gener-
 509 ate enough angular acceleration to reverse the motion. In practical terms,
 510 there is a finite tilt angle within which the rocket can still be brought back
 511 to vertical, and past it the flight becomes unrecoverable. To make this
 512 boundary concrete, the disturbance torque must be written in terms of the
 513 tilt angle. For the finless configuration the dominant disturbance is aero-
 514 dynamic: with the center of pressure ahead of the center of mass, any tilt
 515 produces a destabilizing moment about the center of mass. For a near-
 516 vertical trajectory the tilt angle and the aerodynamic angle of attack are
 517 approximately equal, and at small angles the aerodynamic normal force is

→ is this repeating what was said before?

→ restructure sentence.

→ explain this.

$$N \approx \frac{1}{2} \rho v^2 A_{\text{ref}} C_{N\alpha} \theta$$

where ρ is the air density, v is the airspeed, A_{ref} is the reference cross-sectional area, and $C_{N\alpha}$ is the slope of the normal-force coefficient, obtained from the Barrowman method (5). This force acts at the center of pressure, a distance $\ell = x_{\text{CoP}} - x_{\text{CoM}}$ from the center of mass, so the disturbance torque is

$$\tau_{\text{dist}}(\theta) \approx \frac{1}{2} \rho v^2 A_{\text{ref}} C_{N\alpha} \ell \theta$$

is there a way to relate this to a more simplistic equation?

Setting this equal to the maximum corrective torque from Section 3.4.1 and solving for the tilt gives the maximum recoverable angle:

$$\theta_{\text{max}} \approx \frac{F_T r \sin(\delta_{\text{max}})}{\frac{1}{2} \rho v^2 A_{\text{ref}} C_{N\alpha} \ell} \Rightarrow \text{where } \gamma \text{ is substituted for its previously defined equivalent expression: } F_T r \sin(\delta_{\text{max}}).$$

This form makes the dependence clear: the recoverable tilt rises with motor thrust F_T and falls as airspeed increases, since the aerodynamic disturbance scales with v^2 . The rocket is therefore easiest to recover early in the burn, when thrust is high and airspeed is still low, and progressively harder to recover as thrust decays and dynamic pressure builds.

Evaluated with this vehicle's parameters, the boundary is consistent with the results of simulation and analysis, which indicated that stable recovery was consistently achievable for pitch and yaw deviations below approximately 12° and matched the 12° – 15° envelope mapped numerically in Section 6.10. This value is slightly larger than the mechanical gimbal limit of $\pm 10^\circ$ because recovery depends on angular acceleration applied over time, not on deflection angle alone.

*clunky sentence
⇒ clarify.*

3.4.3 Dynamic Effects and Control Latency

Beyond the static torque limit, tilt recovery is also constrained by dynamic effects: control latency, actuator response speed, and sensor update rate. Even when sufficient corrective torque exists in principle, delays in any of these can allow an angular deviation to grow past the recoverable limit before the correction is fully applied. In an ideal system this delay would be zero, but in practice even a few milliseconds of lag can cause the controller to overcorrect or undercorrect.

→ This means that due to the impacts of dynamic effects and control latency, a measured state where a tilt is theoretically recoverable could propagate into one which is not by the time the corrective action is applied.

The MG90S servos have a finite response time of roughly 0.12 s per 60° of motion, which limits how quickly the thrust vector can be redirected against a sudden disturbance. If the rocket experiences a rapid tip-off immediately after launch, as lightweight vehicles often do, the servo response can lag the disturbance, allowing angular velocity to build faster than the system can arrest it. Sensor processing adds a smaller delay of its own: although the IMU reports data frequently, the Kalman filter smooths rapid jumps in that data to reject noise, which introduces a small but unavoidable ~~phase~~ lag in the attitude estimate. This trade-off improves stability under vibration at the cost of some responsiveness to very fast disturbances.

→ a trait that lightweight vehicles are susceptible to

→ oppose it.

→ rephrase because wording is weird.
↳ use rapid?

Together, these effects impose a dynamic recovery limit, one where recovery fails not because the available torque is insufficient, but because the correction arrives too late. Two design choices mitigate this risk: selecting conservative PID gains that prioritize stability and predictability over aggressive correction, and running the control loop well above the actuator bandwidth to minimize computational delay and issue corrective commands as early as possible. Taken together, the static and dynamic limits define a specific, calculable domain of controllability within which stable flight and recovery are possible. Characterizing this domain was essential both to the mechanical design of the gimbal and to the tuning of

→ restructure without colon → not my voice.

565 the closed-loop control algorithm.

→ explain how choosing characteristics of the control system depends on these limitations.

OVERALL, SOLID SECTION → VERY THOROUGH.

566 4 Flight Software and Autonomous State Transitions

567 The flight software serves as the layer that governs when the control system is enabled, how sensor data is interpreted, and how the system transitions between distinct phases of flight. While the thrust vector control system defines how correction actions are computed, the flight software determines when such corrections are physically necessary and safe to apply. For this project, an autonomous state-machine-based architecture was implemented to ensure data-backed behavior, robustness that protects against sensor noise, and safe transitions between powered and unpowered flight. The software executes on a Teensy 4.1 microcontroller as a high-frequency loop (~ 200 Hz). Within this loop, the PD/PID computation and Kalman filter update specifically execute at 150 Hz, matching the control update rate used in the validation simulations of Section 6, while sensor sampling, state-machine evaluation, and data logging proceed at the full loop rate. → of 200Hz?

582 4.1 Autonomous Staging and Flight Phase Identification

583 At the core of the flight software lies the autonomous staging system, responsible for identifying the rocket's current phase of flight based on real-time sensor data. This system evaluates accelerometer, gyroscope, and derived kinematic quantities to determine whether the rocket is idle, undergoing powered ascent, coasting, descending, or has completed flight. Each flight phase corresponds to a unique control context, ensuring that thrust vector corrections are applied only when thrust is available and that

→ such as what?

→ This sentence works but I probably wouldn't write it this way.

590 recovery logic is activated at the appropriate moment. Rather than relying
591 on a single sensor or threshold, the staging logic evaluates multiple indi-
592 cators simultaneously. This multi-parameter approach improves robust-
593 ness against noise, vibration, and transient disturbances, particularly dur-
594 ing high-dynamic events such as launch and motor burnout. The overall
595 staging logic follows the structure illustrated in Figure A1, which outlines
596 the transitions between flight states.

Is this design choice necessary to justify or not?

597 4.2 State Machine Logic and Execution Flow

598 The flight software is structured as a state machine operating within a con-
599 tinuous control loop. During each iteration of the loop, current sensor
600 measurements are processed and compared against state-dependent transi-
601 tion criteria. The system begins in a pad idle state, during which the rocket
602 is stationary and awaiting launch. In this state, sensor calibration is com-
603 pleted, baseline offsets are established, and all actuators are held in neutral
604 positions to prevent unintended motion.

→ flight state

→ new paragraph
& add sentence
before that says
something like: the
following is how the
computer operates.

605 Upon detection of launch conditions, the system transitions into the pow-
606 ered flight state. In this phase, thrust vector control is fully enabled, and
607 the PD controller actively stabilizes pitch and yaw using Kalman-filtered
608 attitude estimates. The powered flight state persists until motor burnout
609 and upward velocity decay are detected, at which point the software tran-
610 sitions into an apogee detection state. This intermediate state allows the
611 system to distinguish between transient velocity fluctuations and the true
612 apex of the trajectory.

→ weird wording.

↳ Also explain why
apex was not determined
by 0 velocity because
the sensors may not
always pick up 0 velocity
→ emphasize robustness
of control system.

613 Once apogee is confirmed, the system enters the descent state. During de-
614 scent, active thrust vector control is disabled, as thrust is no longer avail-
615 able to generate corrective torque. The final transition occurs after landing
616 is detected, at which point the system enters a recovery or post-landing

↓
covered in section 4.5
⇒ make this clear.

617 state where data logging is finalized and active control subsystems are
618 powered down.

619 This state-based execution model ensures that control logic is always context-
620 aware and prevents undesired actuation when it is not required or will not
621 have a meaningful effect on controlling the rocket's trajectory.

622 4.3 Sensor Inputs, Calibration, and Filtering

623 Reliable autonomous decision-making depends on accurate and stable sen-
624 sor data. The rocket employs a six-degree-of-freedom MPU-6050 iner-
625 tial measurement unit, providing three-axis accelerometer and gyroscope
626 measurements. While these sensors offer high sampling rates and low la-
627 tency, their raw outputs are susceptible to bias, drift, and vibration-induced
628 noise. Gyroscopes measure angular velocity and therefore accumulate in-
629 tegration error over time, leading to gradual drift in estimated orientation.
630 Accelerometers, while capable of providing absolute tilt references, are
631 sensitive to high-frequency vibration and non-gravitational accelerations
632 during powered flight. To address these limitations, sensor calibration
633 routines can be performed prior to launch to remove static offsets and real-
634 time filtering is applied during flight. Filtered sensor data is then passed
635 to the Kalman filter described in Section 3.2, producing a smooth and re-
636 liable estimate of the rocket's orientation. This filtered attitude estimate
637 is used both for control input during powered flight and for flight state
638 determination throughout the mission.

*This whole section
is well written
but feels slightly
repetitive
bc context is covered
beforehand.*

*→ through the kalman
filter*

639 4.4 Launch Detection Criteria

640 Accurate launch detection is critical for enabling thrust vector control at
641 the correct moment. Premature activation can lead to unnecessary servo
642 actuation on the launch rail, while delayed activation reduces the system's

ability to counter early disturbances. To balance responsiveness and robustness, launch detection is based on multiple simultaneous conditions rather than a single acceleration threshold.

The transition from pad idle to powered flight is triggered only when sustained axial acceleration exceeding a predefined threshold is detected, accompanied by a consistent increase in vertical velocity over a short time window. Additionally, a minimum arming time must elapse to prevent accidental activation during handling, and gyroscope noise must remain below a defined threshold to ensure a stable sensor environment. Together, these conditions significantly reduce the probability of false positives while ensuring that thrust vector control engages directly after liftoff.

→ explain what occurs if the rocket (because it is so lightweight) bllts directly off the launchpad.

4.5 Apogee Detection Algorithm

Apogee detection represents a critical transition between ascent and descent logic, particularly in thrust-vector-controlled rockets where trajectory deviations can introduce oscillatory velocity behavior. Relying solely on instantaneous vertical velocity reaching zero is unreliable in the presence of noise and transient disturbances. To improve robustness, a multi-condition apogee detection algorithm was implemented.

Apogee is identified only when negative vertical velocity is sustained over a defined time interval and when altitude is no longer increasing, as determined by the derivative of altitude data. A minimum time since launch is also enforced to prevent false detections during early flight transients. Additionally, pitch and yaw deviations must remain within acceptable bounds to reduce the likelihood of erroneous state transitions during unstable flight.

Once all conditions are satisfied, the system transitions into the descent phase and prepares recovery sequencing. As a safeguard, a timeout mecha-

→ does the flight computer have a barometer or some sort of other way to measure altitude since it does not have a barometer. Also explain why a barometer was excluded from this design (reliance on air pressure measurements).

nism forces the transition if apogee is not detected within a predefined time window, ensuring that recovery logic is not indefinitely delayed. Apogee detection events are logged to onboard storage and indicated through diagnostic outputs to allow post-flight validation.

4.6 Descent and Recovery Phase

Following apogee detection, the rocket enters the descent phase. During this phase, thrust vector control is fully disabled and the gimbal is mechanically locked, as no thrust is available to generate corrective torque. Sensor monitoring continues, allowing the system to track altitude and vertical velocity for landing detection and post-flight analysis.

Recovery mode is initiated once the rocket descends below a safe altitude threshold and vertical velocity remains sufficiently low for a sustained period, indicating ground contact or near-ground conditions. At this point, final flight data is logged, memory storage is secured, and power to active control components is shut down. The system then enters a passive idle recovery state, awaiting retrieval.

→ define how it is measured again.

This approach prevents post-landing data corruption and protects actuators from damage due to unnecessary motion after touchdown.

4.7 Abort Mechanisms and Fail-Safe Logic

To ensure system safety and data integrity, multiple fail-safe mechanisms are embedded within the flight software. These safeguards monitor for abnormal servo behavior, sensor faults, and missed state transitions. In the event of excessive servo command rates, thrust vector control is immediately disabled and the fault is logged. If apogee is not detected within the expected time window, recovery mode is forced to prevent indefinite powered logic execution.

→ to the onboard SD card.

695 Manual abort capability is also provided during pre-launch conditions, al-
696 lowing the system to return safely to a standby state in the event of sensor
697 anomalies or excessive pre-launch tilt. In the case of system crashes or un-
698 expected resets, periodic non-volatile memory backups ensure that partial
699 flight data can still be recovered.

explain how.
would this be
through a telemetry
downlink?

→ define necessity.

700 4.8 Future Enhancements to Flight Logic

701 While the current flight software architecture is able to demonstrate reli-
702 able performance, several enhancements could further improve robustness
703 and accuracy. Integrating a high-resolution barometric pressure sensor
704 would provide an independent altitude measurement for improved apogee
705 detection, though care must be taken to mitigate pressure disturbances
706 caused by airflow. Adaptive control strategies, such as in-flight PID gain
707 adjustment or model predictive control, could further optimize stabiliza-
708 tion performance under varying flight conditions. Additionally, extending
709 the state machine to support multi-stage propulsion or stage separation de-
710 tection would enable more complex mission profiles.

→ Alright, BMP is
mentioned here.

→ explain why this
would be useful.

→ change phrasing → avoid 'mission
profiles'.

711 5 Simulations and Performance Modeling

712 Simulation and modeling played a central role in validating the feasibil-
713 ity of the thrust vector control system prior to experimental flight testing.
714 Given the risks associated with uncontrolled instability in TVC rockets,
715 analytical and numerical models were used to predict vehicle behavior un-
716 der both ideal and disturbed conditions, establishing performance expec-
717 tations against which experimental data could later be compared. Rather
718 than relying on a single framework, a hybrid modeling approach was used:
719 OpenRocket (4) analyzed baseline trajectory behavior and aerodynamic
720 stability using established, Barrowman-based rocketry methods (5), while

→ explain what
and why.

721 custom analytical models evaluated thrust vector control authority, angu-
722 lar response, and recovery limits. This separation allowed aerodynamic
723 and control effects to be studied independently before being treated as a
724 coupled system. The OpenRocket baseline reproduced the geometry, mass
725 distribution, and Estes E12-4 thrust curve of Section 2 and confirmed the
726 static instability anticipated in Section 2.5: the center of pressure remained
727 ahead of the center of mass, producing a negative static margin and rein-
728 forcing the necessity of active stabilization.

→ good, but
rewrite sentence
because of tone
and use of colon.

→ check these
references.

729 Two qualitative findings from this modeling stage directly shaped the de-
730 sign and are quantified in the validation results of Section 6. First, using
731 the corrective-torque relationship of Sections 3.3 and 3.4.1, the maximum
732 available control authority was found to be highest shortly after ignition
733 and to decay steadily toward burnout; this informed the launch-detection
734 and apogee logic of Section 4, which concentrates control effort during
735 the early powered phase when both thrust and corrective torque are great-
736 est. Second, a rigid-body tip-off model confirmed that small initial tilts are
737 arrested within a fraction of a second, whereas deviations beyond a crit-
738 ical angle diverge, consistent with the recovery limits derived in Section
739 3.4 and justifying the conservative launch thresholds implemented in the
740 flight software.

→ Rewrite entirely
→ tone is not
like me at all.

741 Ultimately, these simulations were not intended to predict flight behavior
742 exactly, but to define a realistic operating envelope for the thrust vector
743 control system. By identifying control authority limits, disturbance sensi-
744 tivity, and recovery boundaries in advance, they reduced the risk of catas-
745 trophic failure during experimental testing.

→ explain why this
would be incredibly
difficult to model.

→ add: which would take weeks/months to rebuild the
rocket

6 Simulation-Based Validation

Due to airspace constraints, no powered flight testing was conducted. System behavior was evaluated using a physics-based simulation incorporating rigid-body rotational dynamics, thrust vector geometry, actuator limits, and PID control laws.

6.1 Overview of Simulation Objectives

To evaluate the performance and robustness of the thrust vector control system in the absence of powered flight testing, a comprehensive set of physics-based simulations was developed in Python. The simulation integrates rigid-body rotational dynamics using a fourth-order Runge-Kutta (RK4) scheme at a 1 ms time step, with the Estes E12-4 thrust curve represented by NAR-certified data points interpolated via numpy. The control architecture, which includes the discrete-time PD controller (the integral term is disabled for this simulation, $K_i = 0$, per Section 3.1), Kalman filter, and servo rate limiting, runs at 150 Hz within the simulation loop to match the intended flight computer update rate. It is important to note that the simulations assume a rigid airframe, constant atmospheric conditions, and no launch rail dynamics. These simplifications are discussed further in Section 6.12.

→ clarify better.

→ and in the Appendix maybe?

6.2 Motor Thrust Profile

Before examining control system behavior, it is useful to understand the propulsive input driving the simulation. Figure 1 shows the NAR-certified thrust curve for the Estes E12-4 motor, interpolated from published data points sourced from ThrustCurve.org (9). The motor produces a sharp peak of 33.0 N at approximately $t = 0.29$ s, followed by a gradual decay to a plateau of roughly 9.5 N before burnout at 2.44 s, with a mean thrust of

11.12 N. This time-varying thrust profile is significant for control system design: available corrective torque is directly proportional to instantaneous thrust, meaning control authority is highest immediately after ignition and diminishes steadily toward burnout.

→ discuss implications of this and take color out of sentence.

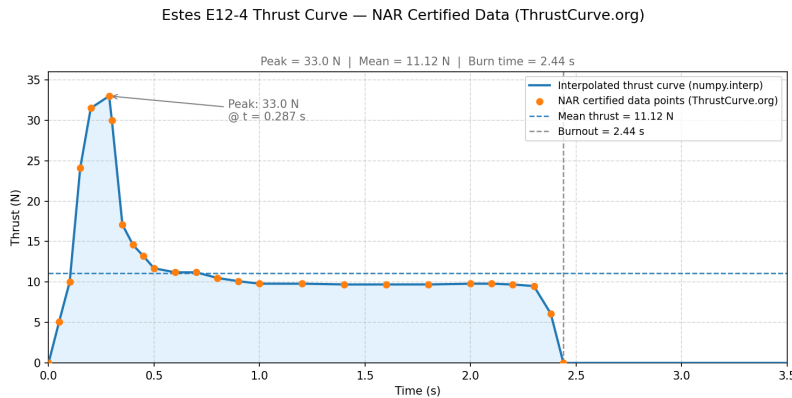


Figure 1: Estes E12-4 Thrust Curve — NAR Certified Data (ThrustCurve.org). Peak = 33.0 N at $t = 0.29$ s, Mean = 11.12 N, Burn time = 2.44 s.

6.3 Baseline Closed-Loop Response

Figure 2 shows the baseline closed-loop attitude response to a 5° initial pitch offset, representing a realistic launch tip-off disturbance. The top panel shows that the true pitch angle is driven back toward vertical within approximately 0.3 seconds of ignition, with the Kalman filter estimate tracking closely during powered flight. The middle panel shows the corresponding angular rate, which peaks near -45 deg/s immediately after correction begins before decaying smoothly as the controller damps rotational motion. The bottom panel shows gimbal deflection, which briefly approaches the mechanical limit at the onset of correction before reducing to small oscillatory commands as the vehicle settles. After motor burnout at 2.44 s the control loop is disabled and, with the vehicle in near-freefall, the Kalman estimate drifts on gyroscope integration alone; this post-burnout behavior is examined in Section 6.7 and is distinct from the roll-axis, magnetometer-related drift of Appendix A.7.2.

→ as discussed in section 2 as a real possibility for light weight vehicles of this sort.

→ of attitude

fix and clarify this is different than mechanically feasible max gimbal angle.

→ explain again why commanded $\neq$ actual deflection.

↳ rewrite: tone.

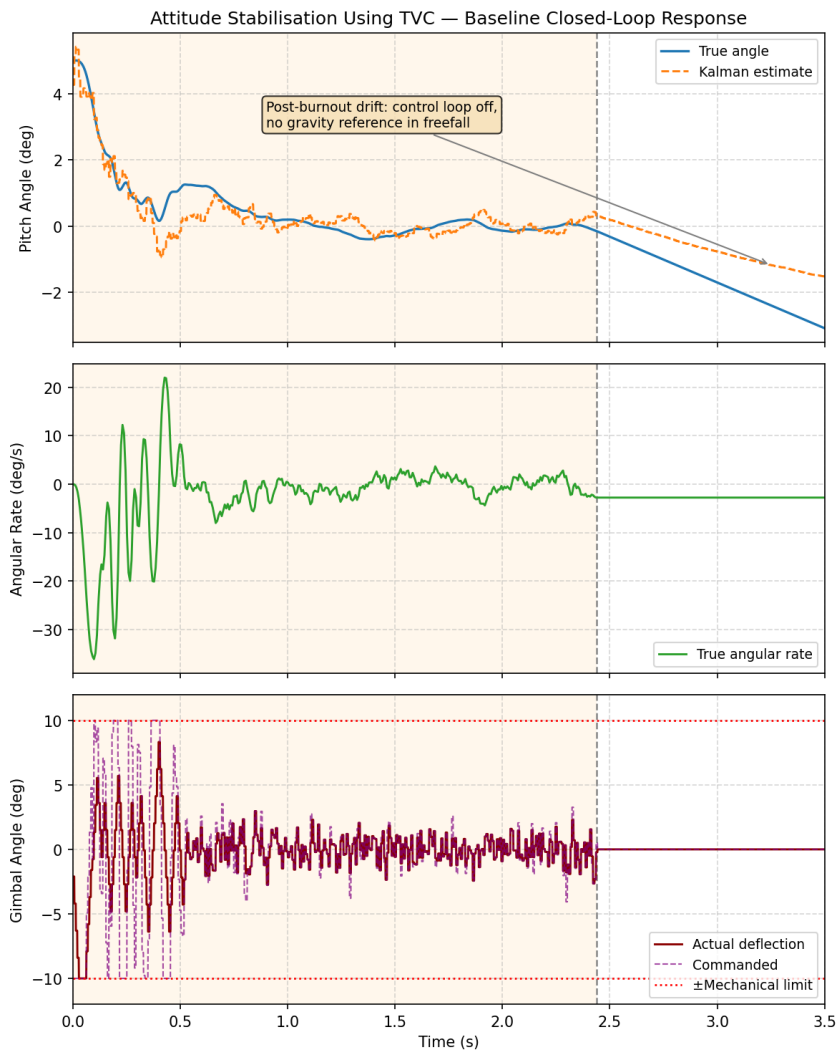


Figure 2: Attitude Stabilisation Using TVC — Baseline Closed-Loop Response. Top: pitch angle (true vs. Kalman estimate). Middle: angular rate. Bottom: gimbal deflection with $\pm 10^\circ$ mechanical limits shown. Motor burn window shaded; post-burnout gyroscopic drift annotated.

6.4 Open-Loop vs. Closed-Loop Comparison

Figure 3 provides the most direct argument for the necessity of active thrust vector control in this finless configuration. Starting from the same 5° initial offset, the open-loop case—representing a rocket with no active stabilization—shows no tendency to return toward vertical. Because the vehicle is aerodynamically unstable in this finless configuration (Section 2.5), there is no restoring moment to correct the deviation; at the relatively low dynamic pressures and short burn duration modeled here, the

This is great but the sentence is very long.

destabilizing aerodynamic moment is not large enough to produce visible growth of the tilt angle over the burn window shown, but the same lack of a restoring moment means that, given more time, a larger initial disturbance, or higher dynamic pressure, this deviation would diverge rather than decay. The closed-loop case corrects the same disturbance within approximately 0.3 seconds and maintains near-zero pitch angle through burnout. This contrast illustrates that passive stability is entirely absent in this design, and that the PID-Kalman control architecture is solely responsible for maintaining a stable trajectory.

and confirms that the system is aerodynamically unstable.

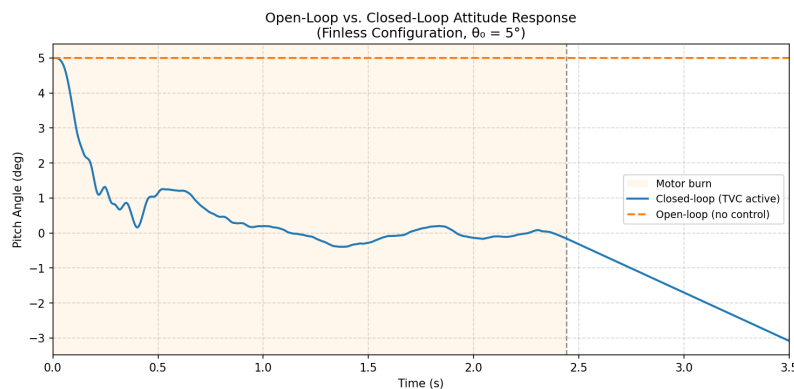


Figure 3: Open-Loop vs. Closed-Loop Attitude Response (Finless Configuration, $\theta_0 = 5^\circ$). Without active control, the initial offset persists throughout the burn. Closed-loop TVC corrects the disturbance within ~ 0.3 s.

6.5 Disturbance Rejection Response

Figure 4 evaluates the system's ability to reject a mid-flight external disturbance. A torque impulse of $0.12 \text{ N}\cdot\text{m}$ applied for 50 ms at $t = 0.6 \text{ s}$ simulates the effect of a wind gust or aerodynamic transient during powered ascent. The disturbance produces a peak angular deviation of approximately 1.3° before the controller arrests the motion and returns the vehicle toward its nominal trajectory. Recovery is stable and completes within 0.4 seconds. This result suggests the system has reasonable tolerance to aerodynamic disturbances within the modeled range.

take out or simplify.

rewrite: tone / author's voice.

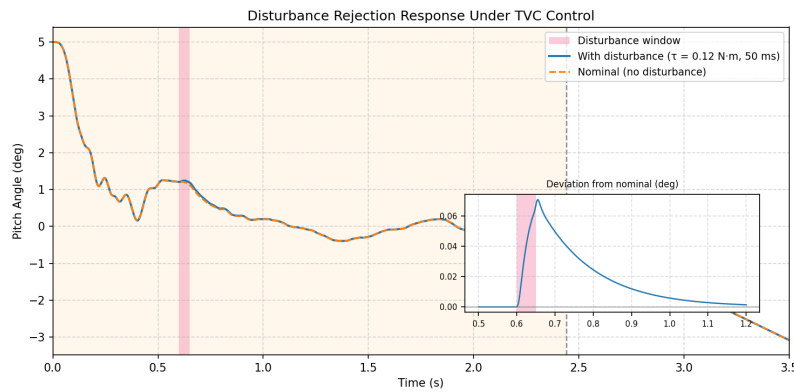


Figure 4: Disturbance Rejection Response Under TVC Control ($\tau = 0.12 \text{ N}\cdot\text{m}$, 50 ms at $t = 0.6 \text{ s}$). Disturbance window highlighted. Nominal (no-disturbance) trace shown for comparison.

6.6 Control Effort and Actuator Utilization

Figure 5 shows how the controller converts attitude errors into actuator movement and confirms that the servo system remains within its limits. In the top panel, a slight lag between commanded and actual gimbal deflection highlights the servo's rate limiting during the initial correction. The bottom panel shows that cumulative actuator saturation is confined almost entirely to the initial tip-over correction ($t < 0.06 \text{ s}$), during which the commanded signal briefly reaches the $\pm 10^\circ$ mechanical limit; the cumulative saturated fraction decays rapidly thereafter and settles at approximately 1.0% of the total burn. The top panel's slight lag between the commanded signal and the rate-limited actual deflection is the servo physically catching up to this brief initial command. Together these confirm that the PD gains are well-tuned for the MG90S servo: saturation is limited to the brief initial correction rather than persisting throughout the burn, and the system maintains stability without excessive sustained actuator demand.

written different
than the rest of
the paper
 $\Rightarrow$ match writing
style.

Explain why cumulative
actuator saturation is a good
thing to measure and what it
might mean for future work
(especially on the mechanics of the
gimbal).

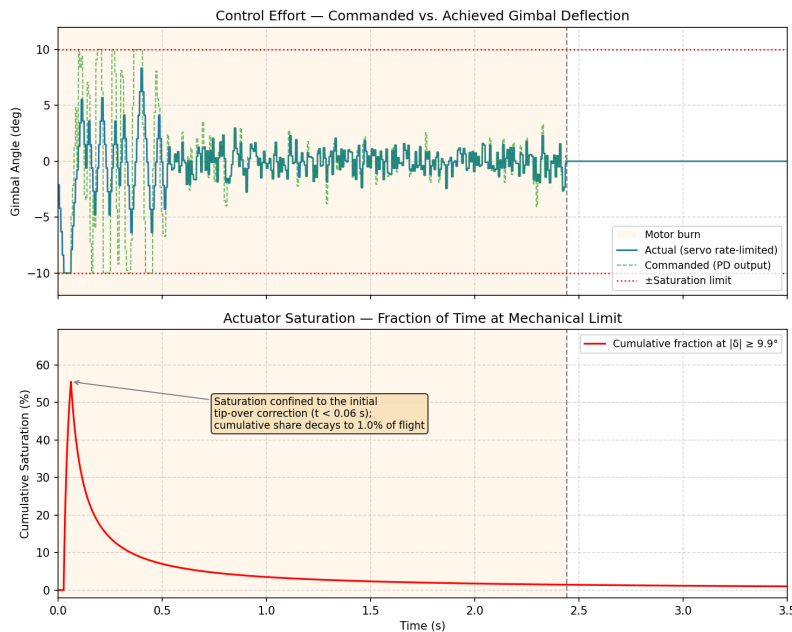


Figure 5: Control Effort — Commanded vs. Achieved Gimbal Deflection (top) and Cumulative Actuator Saturation as a fraction of burn time (bottom). Saturation is confined to the initial tip-over correction ($t < 0.06$ s) and decays to approximately 1.0% of the burn thereafter, confirming gains are appropriate for the MG90S servo.

rewrite

6.7 Kalman Filter Performance

Figure 6 isolates the performance of the Kalman filter by comparing the raw accelerometer signal, the filtered attitude estimate, and the noise-free ground truth across the full simulation window. During powered flight, the raw accelerometer output is heavily corrupted by vibration-induced noise, with instantaneous errors exceeding $\pm 4^\circ$. The Kalman estimate tracks the ground truth closely throughout the burn, suppressing high-frequency noise while preserving the low-frequency attitude trend. After burnout, the filtered estimate begins to diverge from ground truth: with the control loop disabled and the vehicle in freefall, the accelerometer no longer senses a usable gravity vector, so the filter falls back on gyroscope-only integration, which drifts over time. This is a separate effect from the roll-axis heading drift discussed in Appendix A.7.2, which stems from the MPU-6050's lack of a magnetometer (8) and applies only during powered flight. This post-

as mentioned in section...

burnout behavior is expected and does not affect the control system, which is disabled after motor shutdown.

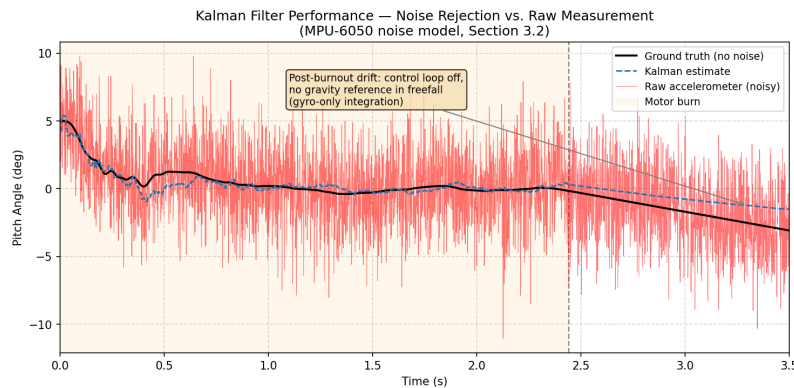


Figure 6: Kalman Filter Performance — Noise Rejection vs. Raw Measurement (MPU-6050 noise model, Section 3.2). The filter suppresses $\pm 4^\circ$ peak accelerometer noise during powered flight. Gyroscopic drift post-burnout is annotated; it does not affect closed-loop stability.

6.8 TVC Corrective Torque Authority

Figure 7 plots the maximum corrective torque available from the TVC system throughout the motor burn, calculated using the relationship from Section 3.4.1:

$$\tau_{\max} = F_T \cdot r \cdot \sin(\delta_{\max})$$

The torque peaks at 1.65 N·m near maximum thrust and holds steady at approximately 0.5 N·m during the sustained burn phase. Since the control authority margin remains substantial compared to a representative disturbance torque of 0.025 N·m, stability is maintained throughout powered flight. The corrective authority profile mirrors the thrust curve, as expected from the governing equation. The collapse of available torque at burnout is why the flight software disables active control immediately after motor shutdown.

explain disturbance torque magnitude or omit from paper & figure caption.

and confirms a proportional relationship between corrective authority & available thrust.

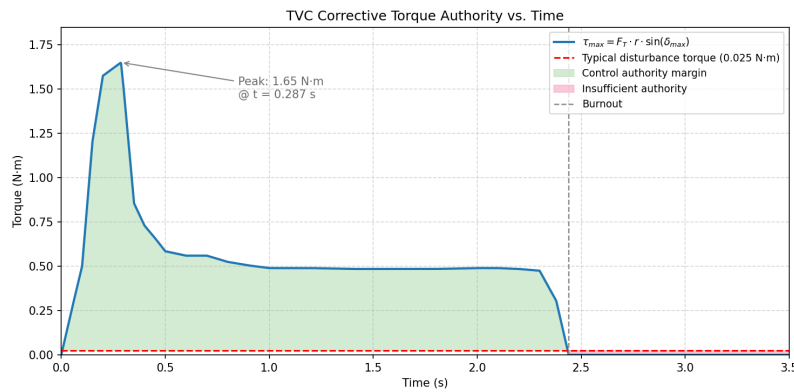


Figure 7: TVC Corrective Torque Authority vs. Time ($\tau_{\max} = F_T \cdot r \cdot \sin(\delta_{\max})$). Green shading: control authority margin above typical disturbance (dashed red). Burnout at 2.44 s marked; torque collapses immediately thereafter.

6.9 Parametric Sensitivity Analysis

Figure 8 analyzes the pitch response sensitivity to three design parameters: motor thrust scaling, CoM-to-nozzle distance, and maximum gimbal angle. Variations in motor thrust ($\pm 30\%$) and CoM-to-nozzle distance ($\pm 20\%$) caused negligible changes in recovery performance, confirming the control architecture's robustness to manufacturing tolerances and motor lot variation. Maximum gimbal angle had the greatest effect: stable recovery was maintained even when the limit was reduced to 6° or 8° , while increasing it to 12° produced a slightly more aggressive initial correction. This confirms that the 10° design choice offers adequate authority with a meaningful margin, and that stability is unlikely to be compromised by minor assembly deviations.

good but it might help to explain how this model was created.

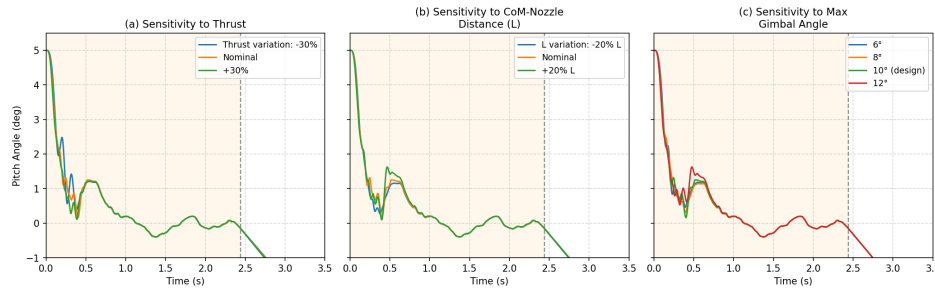


Figure 8: Parametric Sensitivity Analysis — Pitch Response ($\theta_0 = 5^\circ$). (a) Motor thrust variation ($\pm 30\%$). (b) CoM-to-nozzle distance variation ($\pm 20\%$). (c) Maximum gimbal angle (6° , 8° , 10° design, 12°).

6.10 Controllability Boundary

Figure 9 maps the two-dimensional space of initial pitch angle and initial angular rate into recoverable and unrecoverable regions, providing a visual representation of the domain of controllability discussed analytically in Section 3.4. The green region indicates initial conditions from which the controller successfully recovers the vehicle during powered flight; the red region represents conditions from which divergence is unavoidable given the gimbal's physical limits. The boundary occurs at approximately 12° – 15° of initial pitch angle depending on initial angular rate, consistent with the theoretical estimate derived in Section 3.4.2. This figure directly justifies the conservative launch detection thresholds implemented in Section 4: if the vehicle is tilted beyond this boundary at ignition, no amount of control effort can recover stable flight.

→ rephrase.

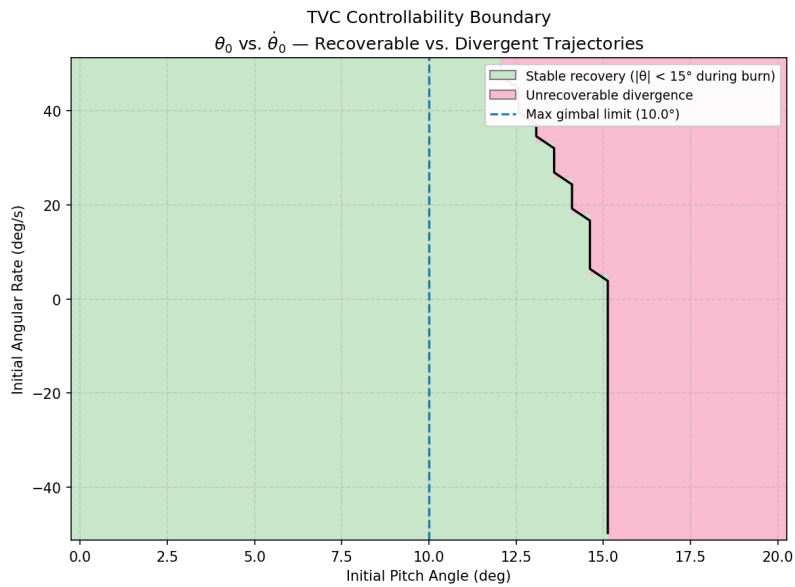


Figure 9: TVC Controllability Boundary: θ_0 vs. $\dot{\theta}_0$ — Recoverable vs. Divergent Trajectories. Green: stable recovery ($|\theta| < 15^\circ$ during burn). Red: unrecoverable divergence. Dashed vertical: max gimbal limit (10.0°).

rewrite.

6.11 Two-Axis Monte Carlo Simulation

Figure 10 provides the section's most thorough validation through 100 independent trials. These trials randomized initial pitch and yaw angles (0.5° to 6°), scaled thrust by $\pm 8\%$ to mimic motor manufacturing variance, and included IMU sensor noise at levels consistent with the MPU-6050 datasheet (8). In every trial, the rocket recovered successfully during powered flight and maintained stable orientation. The rapid narrowing of the 5th-to-95th percentile envelope after the initial correction confirms consistent performance across all input variables. While individual traces diverge after burnout due to independent gyroscopic drift, the results collectively demonstrate that the control architecture is robust against realistic launch variability, provided the vehicle begins flight within the recoverable envelope defined in Section 6.10.

↳ good and clear but rewrite for tone.

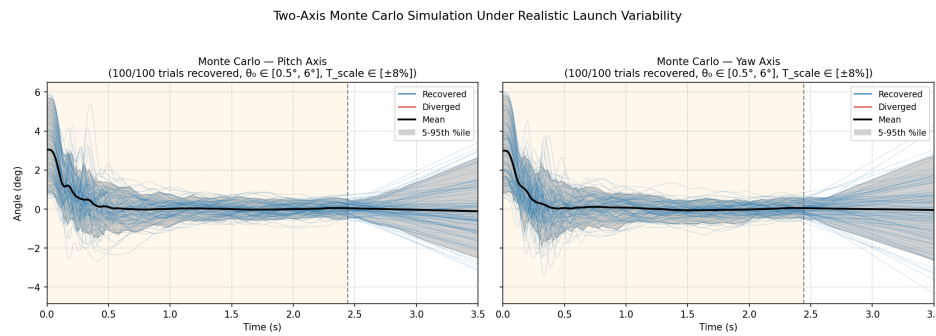


Figure 10: Two-Axis Monte Carlo Simulation Under Realistic Launch Variability (100 trials; $\theta_0 \in [0.5^\circ, 6^\circ]$, $T_{\text{scale}} \in [\pm 8\%]$, sensor noise included). Pitch (left) and yaw (right) axes shown independently. 100/100 trials recovered. Mean and 5th–95th percentile envelope shown.

→ rewrite.

6.12 Discussion of Results and Limitations

Taken together, the simulations demonstrate consistent closed-loop stability across a wide range of initial conditions, disturbance magnitudes, and parameter variations. The control system respects actuator constraints, responds predictably to external inputs, and produces failure modes that align with the analytical limits derived in Section 3.4 rather than arising from control instability. The 100% Monte Carlo recovery rate within the tested envelope provides meaningful confidence in the architecture’s robustness.

The simulations do not capture all real-world effects. Aerodynamic nonlinearities at high angles of attack, structural vibration and aeroelastic coupling, and launch rail dynamics are not modeled. Sensor noise models, while representative of the MPU-6050’s published characteristics (8), cannot fully replicate the high-acceleration environment of an actual motor firing. These results should therefore be interpreted as validation of the system architecture and control design rather than a guarantee of flight performance. Physical validation using a 2-DOF static test stand—described as future work in Section 7—will provide the next level of verification.

→ define.

Table 1: Summary of Simulation-Based Performance Results

Metric	Value
Settling time, baseline 5° tip-off (Section 6.3)	~0.3 s
Peak angular deviation, mid-flight disturbance (Section 6.5)	1.3°
Recovery time, mid-flight disturbance (Section 6.5)	0.4 s
Peak corrective torque, τ_{max} (Section 6.8)	1.65 N·m
Sustained corrective torque, mid-burn (Section 6.8)	~0.5 N·m
Representative disturbance torque (Section 6.8)	0.025 N·m
Cumulative actuator saturation (Section 6.6)	~1.0% of burn
Maximum recoverable tilt angle, θ_{max} (Sections 3.4.2, 6.10)	12°–15°
Kalman filter noise reduction vs. raw accelerometer (Section 3.2)	70–80%
Monte Carlo recovery rate, 100 trials (Section 6.11)	100/100

6.13 Reproducibility and Data Availability

All simulation scripts, parameter files, and plotting code used to generate the figures in this section are available in the public GitHub repository referenced in Appendix A.6. Providing access to these materials allows the results to be independently verified and enables future refinement of the model.

7 Conclusion

This project began with a straightforward question: could a high school student, working independently, design and implement a functional thrust vector control system for a model rocket from the ground up? The answer, after years of iterative hardware design, self-directed learning, and rigorous analysis, is yes. The process of getting there is perhaps as significant as the result itself.

From a technical standpoint, this paper has demonstrated that thrust vector control is a viable stabilization mechanism at the hobby scale. The PID-Kalman control architecture, implemented on a Teensy 4.1 microcontroller and driven by real-time MPU-6050 sensor data (8, 10), produced

↳ reference to what?

933 stable closed-loop behavior across a range of simulated disturbance sce-
934 narios. Simulation results consistently showed that the system could arrest
935 angular deviations, reject external disturbances, and operate within actu-
936 ator limits without passive aerodynamic stabilization. The Monte Carlo
937 analysis further validated the robustness of the control architecture under
938 compounded uncertainty, demonstrating that failure modes were governed
939 by physical constraints rather than control instability.

→ prerequisite for bone.

940 It is important to be clear about the boundaries of these conclusions. The
941 validation presented here is entirely simulation-based. Airspace restric-
942 tions prevented powered flight testing, which means that real-world phenomena—
943 including structural vibration, aeroelastic coupling, launch rail dynamics,
944 and high-acceleration sensor behavior—remain unverified. The rigid-body
945 assumption underlying all simulations is a meaningful simplification, and
946 the gap between simulated and actual flight performance is something only
947 a real launch can close. This paper should therefore be read as a rigorous
948 design and analysis study rather than a flight-validated system.

→ significant simplification.

949 Several natural next steps follow from this work. A companion research
950 study is currently underway to directly quantify the accuracy of the simu-
951 lation framework developed here, using a custom 2-DOF static test stand
952 to conduct live motor firings and compare measured angular error against
953 simulation predictions using the Integral of Absolute Error as the primary
954 performance metric. This work will identify where the simulation di-
955 verges from physical reality and trace that gap back to specific unmodeled
956 phenomena such as actuator rate saturation, gimbal friction, and vibra-
957 tion noise. Beyond that, integrating a barometric pressure sensor would
958 provide an independent altitude measurement for more robust apogee de-
959 tection. On the control side, adaptive gain scheduling—where PID pa-
960 rameters shift as a function of thrust magnitude or estimated dynamic

→ define.

→ through analysis of collected data.

961 pressure—could meaningfully improve performance during the thrust de-
962 cay phase where control authority diminishes most rapidly. Extending the
963 platform to a multi-stage configuration would introduce additional com-
964 plexity in both the state machine and the control architecture, representing
965 a logical progression of the current design.

→ include a reference
to where this
has been shown to
work elsewhere.

966 More broadly, the goal of this project was never exclusively technical.
967 Thrust vector control remains largely inaccessible in hobbyist and educa-
968 tional rocketry, not because the physics are beyond reach, but because the
969 path from concept to implementation is poorly documented at the small
970 scale. If this paper helps another student shortcut even a fraction of the
971 trial and error that went into this system, it will have done exactly what it
972 was meant to do.

A Flight Software Implementation Details

This appendix provides supporting technical material referenced throughout the paper, including flight software logic, control algorithms, and simulation implementations. These materials are included to improve reproducibility, transparency, and technical completeness while preserving the flow and readability of the main body.

All code presented here was developed specifically for this project and executed on the same computational and hardware platforms described in Sections 2–6.

A.1 State Machine Architecture Overview

Listing 1: High-level autonomous flight state machine.

```
if state == "Pad Idle":
    if accel_z > launch_threshold:
        state = "Powered Flight"

elif state == "Powered Flight":
    if vertical_velocity < 0 and time_since_launch >
        0.75:
        state = "Apogee Detection"

elif state == "Apogee Detection":
    if velocity_magnitude < descent_threshold:
        state = "Descent"

elif state == "Descent":
    if altitude < safe_altitude:
        state = "Recovery"
```

```

1001 elif state == "Recovery":
1002     if impact_detected:
1003         state = "Post-landing"
1004

```

While simplified for clarity, this structure captures the core decision-making framework described in Section 4. Additional safeguards, timeouts, and fault-handling logic are implemented in parallel to prevent unsafe or undefined state transitions.

1009 A.2 Apogee Detection Logic

Apogee detection is implemented using a multi-condition logic approach designed to reduce susceptibility to noise, transient oscillations, and sensor bias. Rather than relying on a single zero-crossing of vertical velocity, the algorithm evaluates sustained trends across multiple sensor channels.

→ rewrite: true.

Listing 2: Multi-condition apogee detection algorithm.

```

1014
1015 if vertical_velocity < negative_threshold for > 150
1016     ms
1017     and altitude_rate < epsilon for > 300 ms
1018     and time_since_launch > 0.75 s
1019     and abs(pitch) < 15 deg and abs(yaw) < 15 deg:
1020         apogee_detected = True
1021

```

A timeout safeguard forces a transition into descent mode if apogee is not detected within a predefined maximum flight time. This ensures recovery logic is still executed in the event of unexpected sensor failure or extreme flight deviation.

A.3 Discrete-Time PID Control Law

The PID control system operates independently on pitch and yaw axes using identical control structures. Each loop consumes filtered attitude estimates from the Kalman filter and outputs a commanded gimbal deflection angle. For the simulation results presented in Section 6, K_i was set to zero, so the simulated controller operated as a PD loop (Section 3.1); the integral term and its associated windup protection, shown below, remain implemented in the flight software for use in other flight regimes.

Listing 3: Discrete-time PID update law implemented on the flight computer.

```
error = desired_angle - measured_angle
integral += error * dt
derivative = (error - previous_error) / dt

output = Kp * error + Ki * integral + Kd *
    derivative
output = constrain(output, -max_deflection,
    max_deflection)

previous_error = error
```

Integral windup protection is enforced by limiting the accumulated integral term when the actuator saturates. This prevents excessive overshoot during recovery from large disturbances, as discussed in Section 3.4.

A.4 Kalman Filter for Attitude Estimation

Attitude estimation is performed using a discrete Kalman filter that fuses gyroscope angular velocity data with accelerometer-based tilt estimates. The filter mitigates high-frequency noise from vibration while correcting for long-term gyro drift. The prediction and update equations implemented

▷ explain more clearly.

in software follow the scalar Kalman formulation described in Section 3.2 directly; because each axis (pitch and yaw) is treated as an independent one-dimensional filter, no matrix algebra is required in the implementation.

A.5 Simulation Code and Plot Generation

All simulations presented in Section 6 were generated using Python 3 and the Matplotlib and NumPy libraries. Simulations were executed on a local development environment using Visual Studio Code. The simulation environment assumes a rigid-body model; therefore, structural flex and aeroelastic effects are not factored into the numerical integration.

The following simulation cases are included:

- Motor thrust curve (E12-4, NAR-certified data)
- Baseline closed-loop response (pitch angle, angular rate, gimbal deflection)
- Open-loop vs. closed-loop comparison
- Disturbance rejection response
- Control effort: commanded vs. achieved gimbal deflection
- Kalman filter performance vs. raw accelerometer
- TVC corrective torque authority vs. time
- Parametric sensitivity analysis (thrust, CoM offset, gimbal angle)
- Controllability boundary map
- Two-axis Monte Carlo simulation with randomized initial conditions

Each simulation shares a common structure: (1) definition of vehicle parameters, (2) numerical integration of rotational dynamics via RK4, (3) PID-based control input calculation at 150 Hz, (4) optional disturbance in-

→ The code for each simulation

jection, and (5) data logging and figure generation. Figures are referenced directly in Section 6 and labeled consistently with their corresponding file-names.

A.6 Code Availability and Reproducibility

To support reproducibility and future development, all flight software and simulation code are provided in a public GitHub repository. This repository includes embedded flight software source files, Python simulation scripts, plot generation utilities, and configuration files.

https://github.com/MJDACKIW/Design_and_Validation_of_a_Thrust_Vector_Controlled_Model_Rocket_Using_PID-Kalman_Architecture/tree/main

A.7 Technical Disclaimers and Simulation Assumptions

This section outlines the physical simplifications and hardware constraints assumed during the design, control law development, and simulation of the TVC system. Acknowledging these factors is essential for evaluating the performance envelope and real-world reliability of the vehicle.

A.7.1 Rigid-Body Dynamics Assumption

All simulations in Section 6 treat the rocket airframe as a perfectly rigid body. In practice, aeroelasticity - where the airframe flexes under aerodynamic and inertial loads - can introduce high-frequency noise into IMU data, potentially leading to control-loop instability if the structural resonance frequency approaches the PID sampling frequency. These structural-control interactions are not modeled in the current numerical integration.

→ simplify wording.

A.7.2 Sensor Limitations and Roll Drift

The avionics suite utilizes the MPU-6050 (8), which lacks an onboard magnetometer. Consequently, the Kalman filter provides high-fidelity state estimation for pitch and yaw—the two axes the accelerometer can reference against gravity—but cannot reference an absolute magnetic heading for the roll axis. While the PID algorithm actively stabilizes the vehicle's pitch and yaw, roll (rotation about the rocket's long axis) is subject to uncorrected gyroscopic drift. For the intended short-duration motor burns (2–5 seconds), this drift is considered negligible for maintaining flight stability, since roll has no effect on the vehicle's ability to maintain a vertical trajectory, though it precludes precise heading-based navigation.

simplify wording
→ match tone
of rest of
paper.

A.7.3 Aerodynamic Instability (Finless Configuration)

The decision to omit passive aerodynamic fins renders the vehicle inherently unstable, with the center of pressure (*CoP*) located forward of the center of mass (*CoM*). This creates a continuous inverted-pendulum scenario. The system's stability is entirely dependent on the active control loop and the gimbal's ability to maintain thrust-vector alignment with the vehicle axis. Any failure in the control software or mechanical gimbal response during the boost phase will result in immediate aerodynamic divergence.

A.7.4 Environmental Constants

Simulations assume a standard atmospheric model with constant air density at sea level. Variable wind shear, localized turbulence, and the change in atmospheric pressure with altitude are not factored into the disturbance rejection models. These factors may introduce additional non-linearities during actual flight trials.

?

Although negligible
for a rocket of this
power capacity)

→ wording.

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